

Raghav Govind Jha

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Academic Employment

North Carolina State University ASSISTANT RESEARCH SCHOLAR ↪ with Alexander Kemper, Bojko Bakalov, Yuan Liu	Raleigh, NC, United States 09/2025 - today
Jefferson National Lab (JLab) POSTDOCTORAL RESEARCHER STAFF ↪ with Robert Edwards, Kostas Orginos	Newport News, VA, United States 09/2022 - 09/2025
Perimeter Institute for Theoretical Physics SENIOR POSTDOCTORAL FELLOW ↪ with Pedro Vieira, Guifre Vidal	Waterloo, ON, Canada 09/2019 - 08/2022

Education

PhD Physics, Syracuse University, NY, USA	08/2013 – 05/2019
M.Sc Physics, St. Xavier's College & Bose Institute, Kolkata, India	08/2011 - 05/2013
M.S. in Nanomaterials, Sorbonne Université, Paris, France	09/2010 - 07/2011
B.Sc. Physics (Honors), St. Stephen's College, Delhi, India	07/2007 - 05/2010

Publications

[[Google Scholar](#) 🔗, [INSPIRE](#) 🔗, [ORCID](#) 🔗]

Note: The authors are mostly listed in alphabetical order.

Citation metrics: 650+, h-index: 15, Single-author papers: 4, Published papers as most senior author: 4

44. *Non-stabilizerness and entanglement in (2+1)-dimensional SU(2) lattice gauge theory using tensor networks*
[arXiv:2609.28634](#) 🔗,
Raghav G. Jha, Jaber I. Taher, Muhammad Asaduzzaman, Goksu C. Toga, Bojko Bakalov, Alexander Kemper
43. *Efficient computation of real-time correlators using Pauli Propagation*
[arXiv:2607.24924](#) 🔗, (Submitted to PRX Quantum)
Alexander F. Kemper, **Raghav G. Jha**, Arnab Bachhar, Goksu C. Toga, Mariano Guerrero Perez, Nicholas J. Mayhall
42. *Ground state preparation of random all-to-all Hamiltonians using ADAPT-VQE*
[arXiv:2606.18339](#) 🔗, (Submitted to Phys. Rev. B)
Sabhayata Gupta, Bharath Sambasivam, Sophia E. Economou, Edwin Barnes, Alexander Kemper, **Raghav G. Jha**
41. *Magic and entanglement in 1+1-dimensional SU(2) lattice gauge theory*
[arXiv:2606.09971](#) 🔗, (Submitted to Physical Review Letters)
Raghav G. Jha, Goksu C. Toga, Jaber I. Taher, Bojko Bakalov, Alexander Kemper

40. *Tensor renormalization group approach to the $O(2)$ models via symmetry-twisted partition functions*
[arXiv:2603.02739](#) , [PoS LATTICE2023 \(2025\) 157](#) 
 Shinichiro Akiyama, **Raghav G. Jha**, Jun Maeda, Yuya Tanizaki, Judah Unmuth-Yockey
39. *Quantum simulation of massive Thirring and Gross–Neveu models for arbitrary number of flavors*
[arXiv:2602.22313](#) , [Phys. Rev. D **113**, 114508 \(2026\)](#) 
 Bojko Bakalov, João Getelina, **Raghav G. Jha**, Alexander Kemper, Yuan Liu
38. *Tensor renormalization group approach to critical phenomena via symmetry-twisted partition functions*
[arXiv:2601.02681](#) , [Phys. Rev. D **113**, 074502 \(2026\)](#) 
 Shinichiro Akiyama, **Raghav G. Jha**, Jun Maeda, Yuya Tanizaki, Judah Unmuth-Yockey
37. *Hybrid continuous-discrete-variable quantum computing: a guide to utility*
[arXiv:2511.13882](#)  (Under review)
 A.F. Kemper, A. Alvertis, M. Asaduzzaman, B. N. Bakalov, D. Baron, J. Bierman, B. Burgstahler, S. Chundury, E. R. Das, J. Furches, F. Guo, **Raghav G. Jha**, K. Klymko, A. Kushwaha, A. Li, A. Majumdar, C. O. Marrero, S. Mohapatra, C. Mori, F. Mueller, D. T. Popovici, T. Stavenger, M. Tirfe, N. M. Tubman, M. Zheng, H. Zhou, Y. Liu
36. *On Ising model in magnetic field on the lattice*
[arXiv:2504.18744](#) 
Raghav G. Jha
35. *Finite-temperature phase diagram of the BMN matrix model on the lattice*
[arXiv:2412.13407](#) , [Phys. Rev. D **111**, 094516 \(2025\)](#) 
Raghav G. Jha, Anosh Joseph, David Schaich
34. *Real-time scattering in Ising field theory using matrix product states*
[arXiv:2411.13645](#) , [Phys. Rev. Research **7**, 023266 \(2025\)](#) 
Raghav G. Jha, Ashley Milsted, Dominik Neuenfeld, John Preskill, Pedro Vieira
33. *Quantum computation of $SU(2)$ lattice gauge theory with continuous variables*
[arXiv:2410.14580](#) , [JHEP **06** \(2025\) 084](#) 
 Victor Ale, Nora Bauer, **Raghav G. Jha**, Felix Ringer, George Siopsis
32. *Sparsity dependence of Krylov state complexity in the SYK model*
[arXiv:2407.20569](#) , [Phys. Rev. D **112**, 046017 \(2025\)](#) 
Raghav G. Jha, Ranadeep Roy
31. *Thermal state preparation of the SYK model using a variational quantum algorithm*
[arXiv:2406.15545](#) 
 Jack Araz, **Raghav G. Jha**, Felix Ringer, Bharath Sambasivam
30. *$SU(2)$ principal chiral model with tensor renormalization group on a cubic lattice*
[arXiv:2406.10081](#) , [Phys. Rev. D **110**, 034519 \(2024\)](#) 
 Shinichiro Akiyama, **Raghav G. Jha**, Judah Unmuth-Yockey
29. *Phase diagram of generalized XY model using tensor renormalization group*
[arXiv:2404.17504](#) , [Phys. Rev. D **110**, 034504 \(2024\)](#) 
 Abhishek Samlodia, Vamika Longia, **Raghav G. Jha**, Anosh Joseph
28. *Hamiltonian simulation of minimal holographic sparsified SYK model*
[arXiv:2404.14784](#) , [Nucl. Phys. B **1012** \(2025\) 116815](#) 
Raghav G. Jha

27. *Tensor renormalization group study of 3D principal chiral model*
[arXiv:2312.11649](#) , *PoS LATTICE2023* (2023) 355 
 Shinichiro Akiyama, **Raghav G. Jha**, Judah Unmuth-Yockey
26. *Nonperturbative phase diagram of two-dimensional $\mathcal{N} = (2, 2)$ super-Yang–Mills*
[arXiv:2312.04980](#) , *Phys. Rev. D* **110**, 054507 (2024) 
 Navdeep S. Dhindsa, **Raghav G. Jha**, Anosh Joseph, David Schaich
25. *Sachdev-Ye-Kitaev model on a noisy quantum computer*
[arXiv:2311.17991](#) , *Phys. Rev. D* **109**, 105002 (2024) 
 Muhammad Asaduzzaman, **Raghav G. Jha**, Bharath Sambasivam
24. *Continuous variable quantum computation of the $O(3)$ model in 1+1 dimensions*
[arXiv:2310.12512](#) , *Phys. Rev. A* **109**, 052412 (2024) 
Raghav G. Jha, Felix Ringer, George Siopsis, Shane Thompson
23. *Toward quantum computations of the $O(3)$ model using qumodes*
[arXiv:2308.06946](#) , *PoS LATTICE2023* (2023) 230 
Raghav G. Jha, Felix Ringer, George Siopsis, Shane Thompson
22. *GPU-Acceleration of Tensor Renormalization with PyTorch using CUDA*
[arXiv:2306.00358](#) , *Computer Physics Communications* 294 (2024) 108941 
Raghav G. Jha, Abhishek Samlodia
21. *Notes on Quantum Computation and Information*
[arXiv:2301.09679](#) 
Raghav G. Jha
20. *Supersymmetric Wilson loops on the lattice in the large N limit*
Eur. Phys. J. Spec. Top. 232:355–358 (2023) 
Raghav G. Jha
19. *Non-perturbative phase structure of the bosonic BMN matrix model*
[arXiv:2201.08791](#) , *JHEP* **05** (2022) 169 
 Navdeep S. Dhindsa, **Raghav G. Jha**, Abhishek Samlodia, Anosh Joseph, David Schaich
18. *Thermal phase structure of dimensionally reduced super-Yang–Mills*
[arXiv:2201.03097](#) , *PoS LATTICE2021* (2022) 187 
 David Schaich, **Raghav G. Jha**, Anosh Joseph
17. *Tensor renormalization of three-dimensional Potts model*
[arXiv:2201.01789](#) 
Raghav G. Jha
16. *Introduction to Monte Carlo for Matrix Models*
[arXiv:2111.02410](#) , *SciPost Phys. Lect. Notes* **46** (2022) 
Raghav G. Jha
15. *Large- N limit of two-dimensional Yang–Mills theory with four supercharges*
[arXiv:2109.01001](#) , *PoS LATTICE2022* (2022) 433 
 Navdeep S. Dhindsa, **Raghav G. Jha**, Anosh Joseph, David Schaich
14. *Tensor renormalization group study of the 3d $O(2)$ model*
[arXiv:2105.08066](#) , *Phys. Rev. D* **104**, 094517 (2021) 
 Jacques Bloch, **Raghav G. Jha**, Robert Lohmayer, Maximilian Meister

13. *Three-dimensional super-Yang–Mills theory on the lattice and dual black branes*
[arXiv:2010.00026](#) , *Phys. Rev. D* **102**, 106009 (2020) 
Simon Catterall, Joel Giedt, **Raghav G. Jha**, David Schaich, Toby Wiseman
12. *Positive geometries for all scalar theories from twisted intersection theory*
[arXiv:2006.15359](#) , *Phys. Rev. Research* **2**, 033119 (2020) 
Nikhil Kalyanapuram, **Raghav G. Jha**
11. *Critical analysis of two-dimensional classical XY model*
[arXiv:2004.06314](#) , *J. Stat. Mech.* (2020) 083203 
Raghav G. Jha
10. *Thermal phase structure of a supersymmetric matrix model*
[arXiv:2003.01298](#) , *PoS LATTICE2019* (2020) 069 
David Schaich, **Raghav G. Jha**, Anosh Joseph
9. *Finite N unitary matrix models*
[arXiv:2003.00341](#) 
Raghav G. Jha
8. *Tensor renormalization group study of the non-Abelian Higgs model in two dimensions*
[arXiv:1901.11443](#) , *Phys. Rev. D* **99**, 114507 (2019) 
Alexei Bazavov, Simon Catterall, **Raghav G. Jha**, Judah Unmuth-Yockey
7. *Lattice quantum gravity with scalar fields*
[arXiv:1810.09946](#) , *PoS LATTICE2018* (2019) 043 
Raghav G. Jha, Jack Laiho, Judah Unmuth-Yockey
6. *The properties of D1-branes from lattice super Yang–Mills theory using gauge/gravity duality*
[arXiv:1809.00797](#) , *PoS LATTICE2018* (2019) 308 
Raghav G. Jha
5. *Removal of the trace mode in lattice $\mathcal{N} = 4$ super Yang–Mills theory*
[arXiv:1808.04735](#) , *Phys. Rev. D* **98**, 095017 (2018) 
Simon Catterall, Joel Giedt, **Raghav G. Jha**
4. *Nonperturbative study of dynamical SUSY breaking in $\mathcal{N} = (2, 2)$ Yang–Mills*
[arXiv:1801.00012](#) , *Phys. Rev. D* **97**, 054504 (2018) 
Simon Catterall, **Raghav G. Jha**, Anosh Joseph
3. *Truncation of lattice $\mathcal{N} = 4$ super Yang–Mills*
EPJ Web of Conferences **175**, 11008 (2018) 
Simon Catterall, Joel Giedt, **Raghav G. Jha**
2. *Testing the holographic principle using lattice simulations*
[arXiv:1710.06398](#) , *EPJ Web of Conferences* **175**, 08004 (2018) 
Raghav G. Jha, Simon Catterall, David Schaich, Toby Wiseman
1. *Testing holography using lattice super-Yang–Mills on a 2-torus*
[arXiv:1709.07025](#) , *Phys. Rev. D* **97**, 086020 (2018) 
Simon Catterall, **Raghav G. Jha**, David Schaich, Toby Wiseman

55. *When is a quantum system hard to simulate?* | PHYSICS COLLOQUIUM, NC STATE UNIVERSITY, USA | September 21, 2026 [[Slides](#) [🔗](#)]
54. *Prospects and facets of quantum magic* | TRIANGLE AREA QUANTUM SEMINAR SERIES, NC STATE, USA | September 04, 2026 [[Slides](#) [🔗](#)]
53. *Probing magic and entanglement in 1+1-dimensional $SU(2)$ Hamiltonian lattice gauge theory* | LATTICE 2026, UNIVERSITY OF MARYLAND, MD, USA | July 28, 2026 [[Slides](#) [🔗](#)]
52. *Quantum computation of massive four-fermion models in 1+1- dimensions for arbitrary flavor number* | APS GLOBAL SUMMIT, DENVER, CO, USA | March 18, 2026 [[Slides](#) [🔗](#)]
51. *Classical and quantum computing for problems in physics* | ILLINOIS STATE UNIVERSITY, NORMAL IL, USA | February 02, 2026 [[Slides](#) [🔗](#)]
50. *From qubits and qumodes to quantum fields: quantum information for quantum field theory* | LAWRENCE BERKELEY LAB, BERKELEY, CA, USA | December 01, 2025 [[Slides](#) [🔗](#)]
49. *Quantum computation of random Hamiltonians* | HYBRID CV/DV RETREAT MEETING, FRIDAY INSTITUTE, NC STATE UNIVERSITY, RALEIGH, NC, USA | October 13, 2025
48. *Quantum gravity on noisy quantum computers* | APS GLOBAL SUMMIT, ANAHEIM, CA, USA | March 17, 2025 [[Slides](#) [🔗](#)]
47. *Probing Fundamental Physics in the Age of Quantum Information Processing* | UNIVERSITY OF TENNESSEE KNOXVILLE, USA | March 13, 2025 [[Slides](#) [🔗](#)]
46. *Real-time scattering in Ising field theory* | BROOKHAVEN NATIONAL LABORATORY, UPTON, NY, USA | February 13, 2025 [[Slides](#) [🔗](#)]
45. *Krylov complexity for quantum chaos on quantum computer (KC for QC on QC)* | CFNS WORKSHOP, STONY BROOK UNIVERSITY, NY, USA | February 12, 2025
44. *Scattering in Ising field theory* | UC BERKELEY/LBNL NUCLEAR THEORY SEMINAR, BERKELEY, CA, USA [ONLINE] | January 29, 2025 [[Slides](#) [🔗](#)]
43. *Probing fundamental physics in a new era of computation* | UNIVERSITY OF MIAMI, USA | January 22, 2025 [[Slides](#) [🔗](#)]
42. *State preparation and operator growth of SYK model on IBM quantum computer* | TENSOR NETWORK 2024 WORKSHOP, ISHIKAWA, JAPAN [ONLINE] | November 17, 2024 [[Slides](#) [🔗](#)]
41. *Thermal state preparation and dynamics of random all-to-all fermionic model* | C2QA MEETING, MASSACHUSETTS INSTITUTE OF TECHNOLOGY, BOSTON, USA | July 17, 2024 [[Slides](#) [🔗](#)]
40. *SYK model on a noisy quantum computer - dynamics and state preparation* | MANDELSTAM INSTITUTE FOR THEORETICAL PHYSICS (MITP) AND THE NATIONAL INSTITUTE FOR THEORETICAL AND COMPUTATIONAL SCIENCES (NITheCS), JOHANNESBURG, SOUTH AFRICA [ONLINE] | May 07, 2024 [[YouTube](#) [🔗](#)]
39. *Introduction to tensor networks for classical computing of spin systems and gauge theories* | SET OF FOUR LECTURES AT 14TH JO'BURG SCHOOL ON STRING THEORY AT UNIVERSITY OF PRETORIA, SOUTH AFRICA[ONLINE] | April 29-30 and May 02, 2024 [[YouTube](#) [🔗](#)]
38. *Quantum computing for quantum many-body systems* | WILLIAM & MARY, WILLIAMSBURG, USA | April 17, 2024 [[Slides](#) [🔗](#)]
37. *Approaches to universal quantum computing for spin and gauge models* | UNIVERSITY OF IOWA [ONLINE] | April 16, 2024 [[Slides](#) [🔗](#)]
36. *Random dense Hamiltonians on current noisy quantum computers* | UNIVERSITY OF MARYLAND, USA | March 28, 2024 [[Slides](#) [🔗](#)]
35. *Extracting Physics with IBM's 127-qubit quantum processor* | JEFFERSON LAB, VA, USA | March 13, 2024 [[Slides](#) [🔗](#)]

34. *Real-time dynamics of SYK model on a noisy quantum computer* | WORKSHOP ON ‘TOWARD QUANTUM SIMULATION OF GAUGE/GRAVITY DUALITY AND LATTICE GAUGE THEORY’ | March 05, 2024 [[Slides](#) [↗](#)]
33. *SYK model on a noisy quantum computer* | INDIAN INSTITUTE OF SCIENCE, BANGALORE, INDIA [ONLINE] | February 06, 2024 [[Slides](#) [↗](#)] [[YouTube](#) [↗](#)]
32. *Quantum Computation of the $O(3)$ model using qumodes* | LATTICE 2023 AT FERMILAB, USA | August 02, 2023 [[Slides](#) [↗](#)]
31. *Computation with Quantum Mechanics* | SET OF TWO LECTURES AT QUANTUM COMPUTING BOOTCAMP 2023, JEFFERSON LAB, USA | June 20, 2023 [[Resource](#) [↗](#)]
30. *Can quantum computation improve our understanding of quantum fields?* | SET OF TWO LECTURES AT HUGS 2023 SUMMER SCHOOL, JEFFERSON LAB, USA | June 7, 2023 [[YouTube, Part 1](#) [↗](#)] [[Part 2](#) [↗](#)]
29. *Non-linear sigma models using quantum computation* | C2QA THEORY AND SOFTWARE RETREAT, NEW YORK CITY, USA | May 30, 2023 [[Slides](#) [↗](#)]
28. *Introduction to Quantum Computing methods in Physics* | TATA INSTITUTE, MUMBAI, INDIA [ONLINE] | April 27, 2023 [[Slides](#) [↗](#)] [[YouTube](#) [↗](#)]
27. *Aspects of classical and quantum computing of quantum many-body systems* | ASHOKA UNIVERSITY, SONEPAT, INDIA [ONLINE] | February 10, 2023 [[Slides](#) [↗](#)]
26. *Classical computation using tensor networks and quantum computation with qubits and qumodes* | JEFFERSON LAB, USA | November 14, 2022 [[Slides](#) [↗](#)] [[Video](#) [↗](#)]
25. *Application of tensor methods to real-space renormalization and real-time study of field theories* | BROOKHAVEN NATIONAL LAB (BNL), USA [ONLINE] | October 31, 2022 [[Slides](#) [↗](#)]
24. *New tools for old problems in spin and gauge models on the lattice* | IIT HYDERABAD, INDIA [ONLINE] | October 12, 2022 [[Slides](#) [↗](#)]
23. *Some old problems on the lattice using tensors* | NUMSTRINGS 2022 , ICTS, BANGALORE, INDIA | August 26, 2022 [[YouTube](#) [↗](#)]
22. *Introduction to Quantum Computation using QISKIT* | TWO LECTURES FOR SUMMER SCHOOL 2022 AT RENSSELAER POLYTECHNIC INSTITUTE, USA [ONLINE] | June 22 and 23, 2022 [[Lecture 1 & 2](#) [↗](#)] [[YouTube](#) [↗](#)]
21. *New approach to continuous spin models in two and three dimensions* | NUMERICAL METHODS IN THEORETICAL PHYSICS CONFERENCE, APCTP, POHANG, SOUTH KOREA [ONLINE] | May 17, 2022 [[Slides](#) [↗](#)] [[YouTube](#) [↗](#)]
20. *Holography with large matrices on the lattice* | INSTITUTE OF NUCLEAR SCIENCES, UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO, MEXICO CITY, MEXICO | March 24, 2022 [[Slides](#) [↗](#)]
19. *Large N matrix models using Monte Carlo and Bootstrap* | UNIVERSITY OF SURREY, SURREY, UK [ONLINE] | February 22, 2022 [[Slides](#) [↗](#)]
18. *Introduction to tensor networks and spin systems* | AZIM PREMJI UNIVERSITY, BENGALURU, INDIA | January 11, 2022
17. *Tensor networks and spin models* | INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH (IISER), MOHALI, INDIA | December 7, 2021 [[Slides](#) [↗](#)]
16. *Real-space tensor renormalization for spin models in three dimensions* | PERIMETER INSTITUTE, WATERLOO, CANADA | November 19, 2021
15. *Solving matrix models at large and finite N* | TWO LECTURES FOR SUMMER SCHOOL 2021 AT RENSSELAER POLYTECHNIC INSTITUTE, USA [ONLINE] | June 28 and 29, 2021 [[Lecture 1 & 2](#) [↗](#)]

14. *Holographic gauge theories on the lattice* | DUBLIN INSTITUTE FOR ADVANCED STUDIES, DUBLIN, IRELAND | June 23, 2021 [[Slides](#)] [[YouTube](#)]
13. *Old and new methods for new and old problems in Physics* | INDIAN INSTITUTE OF TECHNOLOGY (IIT) MADRAS, INDIA | March 8, 2021 [[Slides](#)]
12. *Probing holographic dualities with lattice supersymmetric Yang-Mills theories* | MASSACHUSETTS INSTITUTE OF TECHNOLOGY, BOSTON, USA [ONLINE] | February 25, 2021 [[Slides](#)] [[YouTube](#)]
11. *New tool for old problems - Tensor network approach to spin models and gauge theories* | UNIVERSITY OF LIVERPOOL, LIVERPOOL, UK [ONLINE] | October 14, 2020 [[Slides](#)]
10. *Tensor Networks: Algorithm & Applications* | TWO LECTURES FOR CYBERTRAINING SUMMER SCHOOL 2020 AT RENSSELAER POLYTECHNIC INSTITUTE, USA [ONLINE] | June 10 and 11, 2020 [[Lecture 1 & 2](#)] [[Resource](#)] [[YouTube](#)]
9. *Holographic aspects of supersymmetric gauge theories* | PERIMETER INSTITUTE, WATERLOO, CANADA | October 4, 2019
8. *Numerical Approaches to Holography* | ASHOKA UNIVERSITY, SONEPAT, INDIA | August 28, 2019 [[Slides](#)]
7. *Numerical Approaches to Holography* | INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH (IISER) MOHALI, INDIA | August 8, 2019
6. *Holographic dualities and tensor renormalization group study of gauge theories* | PERIMETER INSTITUTE, WATERLOO, CANADA | March 11, 2019 [[Video \(PIRSA, 19030108\)](#)]
5. *Supersymmetry breaking and gauge/gravity duality on the lattice* | UNIVERSITY OF COLORADO BOULDER, COLORADO, USA | April 6, 2018 [[Slides](#)]
4. *Recent results from lattice supersymmetry in $2 \leq d < 4$ dimensions* | ICTS, BANGALORE, INDIA | January 31, 2018 [[YouTube](#)]
3. *Testing gauge/gravity duality using lattice simulations* | CONTRIBUTED TALK AT LATTICE 2017 GRANADA, SPAIN | June 22, 2017 [[Slides](#)]
2. *Testing holography through lattice simulations* | YUKAWA INSTITUTE FOR THEORETICAL PHYSICS, KYOTO, JAPAN | April 4, 2017 [[Slides](#)]
1. *Supersymmetry on the lattice* | APS APRIL MEETING 2016 - SALT LAKE CITY, UTAH, USA | April 17, 2016 [[Slides](#)]

Teaching

1. Recitation Instructor for PHY 216 (General Physics II for Honors and Majors) [[Evaluation report](#)] and Grader for PHY 662 (Quantum Mechanics II) Spring 2019
2. Recitation Instructor for PHY 215 (General Physics I for Honors and Majors) and Grader for PHY 312 (Relativity & Cosmology) Spring 2018
3. Grader for PHY 424 (Electromagnetism) and PHY 360 (Waves and Oscillations) Fall 2016
4. Recitation Instructor for PHY 212 General Physics II Spring 2016
5. Grader for PHY 641 (Statistical Mechanics) and PHY 731 (Electromagnetic theory) Fall 2015
6. Recitation Instructor for PHY 211 General Physics I Fall 2014
7. Recitation Instructor for PHY 211 General Physics I [[Evaluation report](#)] Spring 2014
8. Lab Instructor for PHY 101 General Physics Fall 2013

Awards

1. Henry Levinstein Fellowship for Outstanding Senior Graduate Student - Department of Physics, Syracuse University [USD 2000] 2017
2. College of Arts and Sciences Fellowship for best performance in Graduate Courses - Syracuse University [USD 1700] 2014
3. CSIR/UGC-NET - Junior Research Fellowship (JRF) by Government of India 2013
4. Erasmus Mundus Scholarship for pursuing M.S at Sorbonne Université [EUR 12000] 2010
5. National Top 25 Students (out of 5153 students) in National Graduate Physics Examination (NGPE) conducted by Indian Association of Physics Teachers (IAPT) 2009
6. KVPY (Kishore Vaigyanik Protsahan Yojana) Scholarship by Department of Science & Technology, Government of India [about USD 3500 in two years] 2008
7. Merit certificate by University of Delhi (11th in the university out of ≈ 1200 students) 2008
8. NIUS (National Initiative on Undergraduate Sciences) Fellowship by Tata Institute of Fundamental Research (TIFR), Mumbai 2008

Professional Service and Grants

- Referee for Nature Communications, Nature npj QI (Quantum Information), Physical Review A, Physical Review D, Physical Review Letters, Physical Review Research, Quantum, Sigma, SciPost, European Physical Journal (EPJ), IOP Machine Learning: Science and Technology. *Total papers reviewed:* 18
- Chair of parallel session on ‘Quantum Computation and Information’ at the 43rd Annual Lattice conference at University of Maryland, USA [July 2026]
- Organizer of biweekly hybrid CV/DV meeting at NC State University [Spring 2026]
- Chair of parallel session on ‘Quantum Computation and Information’ at the 40th Annual Lattice conference at Fermilab, USA [August 2023]
- Co-organizer of ‘Quantum Computing Bootcamp’ at Jefferson Lab, USA funded by Quantum Horizons, Department of Energy (DOE) [June, 2023]
- Quantum Fields and Strings Seminar Organizer at Perimeter Institute [January 2020 - March 2021]
- DiRAC computing grant in 2022 for ≈ 24 M core-hours with David Schaich, Toby Wiseman, Anosh Joseph *and* USQCD computing grants of ≈ 12 M core-hours on Fermilab pi0 machine each year in 2017 & 2018

Mentoring Experience

- Nikhil Kalyanapuram (Perimeter Scholar International (PSI) student at Perimeter Institute $\rightarrow$ PhD Penn State $\rightarrow$ Industry) 2019–2020
- Navdeep S. Dhindsa (PhD IISER Mohali $\rightarrow$ Postdoc at Tata Institute, Mumbai) 2020–2023
- Vamika Longia (PhD IISER Mohali) 2021–2022
- Abhishek Samlodia (BS-MS IISER Mohali $\rightarrow$ PhD candidate at Syracuse University) 2021–2024
- Nikhil Bansal (BS-MS IISER Mohali $\rightarrow$ PhD candidate at University of Warwick) 2022–2022
- Shane Thompson (PhD University of Tennessee Knoxville $\rightarrow$ Postdoc at U.S. Naval Research Lab, Washington DC) 2023–2024
- Bharath Sambasivam (PhD Syracuse University $\rightarrow$ Postdoc at Virginia Tech, USA) 2023–2024
- Ranadeep Roy (PhD candidate at The Ohio State University) 2023–2025
- Victor Ale (PhD candidate at University of Tennessee Knoxville) 2024–2025

- Nora Bauer (PhD University of Tennessee Knoxville → IonQ) 2024–2025
- Sabhyata Gupta (PhD candidate at Leibniz Universität Hannover) 2024–2026
- Jaber Ibne Taher (PhD student at NC State University) 2026 -
- Mariano Guerrero Perez (PhD student at NC State University) 2026 -
- Dengyao Luo (PhD student in Applied Math at NC State) 2026 -

Academic References (in alphabetical order)

- SIMON CATTERALL, Professor, Syracuse University, USA [\[smcatter@syr.edu\]](mailto:smcatter@syr.edu) 
- ROBERT EDWARDS, Senior Staff, Theory Center, Jefferson Lab, USA [\[edwards@jlab.org\]](mailto:edwards@jlab.org) 
- ALEXANDER KEMPER, Associate Dean for Research and Professor of Physics, NC State University, Raleigh, USA [\[akemper@ncsu.edu\]](mailto:akemper@ncsu.edu) 
- JOHN PRESKILL, Richard P. Feynman Professor of Theoretical Physics, California Institute of Technology, USA [\[preskill@caltech.edu\]](mailto:preskill@caltech.edu) 
- DAVID SCHAICH, Lecturer in Theoretical Particle Physics, University of Liverpool, Liverpool, UK [\[david.schaich@liverpool.ac.uk\]](mailto:david.schaich@liverpool.ac.uk) 
- GEORGE SIOPSIS, Professor, University of Tennessee Knoxville [\[gsiopsis@utk.edu\]](mailto:gsiopsis@utk.edu) 
- PEDRO VIEIRA, The Clay Riddell Paul Dirac Chair in Theoretical Physics at Perimeter Institute, Canada & ICTP-SAIFR, São Paulo, Brazil [\[pvieira@perimeterinstitute.ca\]](mailto:pvieira@perimeterinstitute.ca) 
- TOBY WISEMAN, Professor of Theoretical Physics, Imperial College, London, UK [\[t.wiseman@imperial.ac.uk\]](mailto:t.wiseman@imperial.ac.uk) 